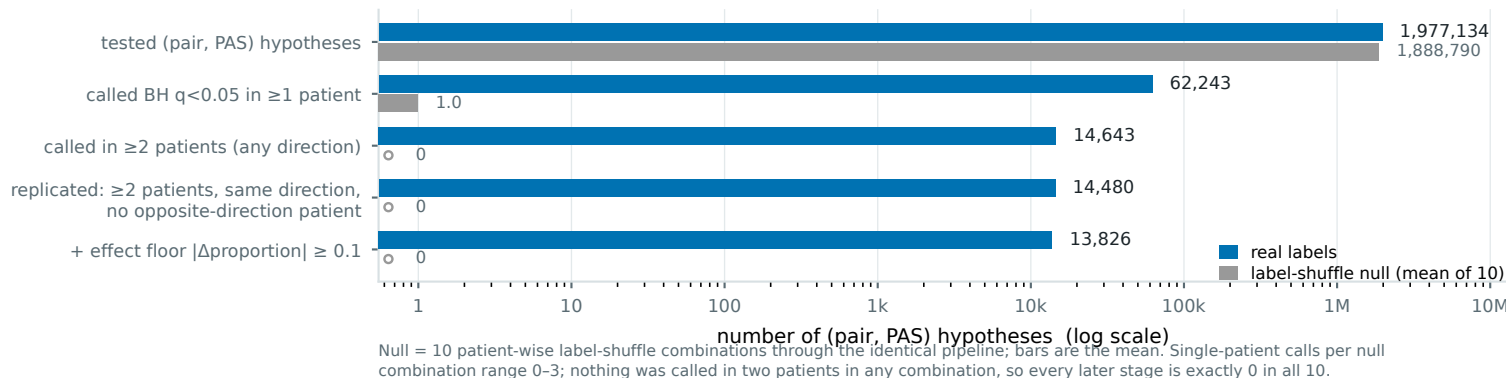


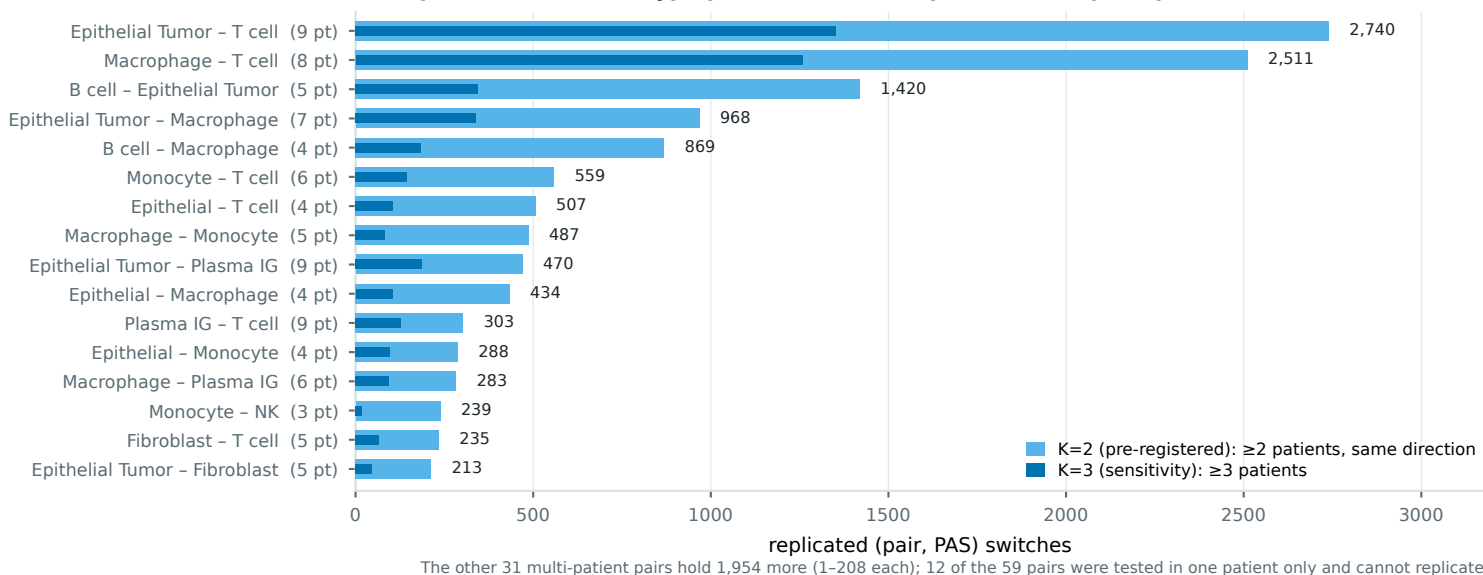
Fig 6 | Reliable cell-type APA switches in a tumour cohort

Laughney lung adenocarcinoma, 12 patients (15 scRNA-seq libraries): 14,480 of 1,977,134 tested (pair, PAS) hypotheses replicate in ≥ 2 patients in the same direction — and none replicate in any of the 10 patient-wise label-shuffle nulls.

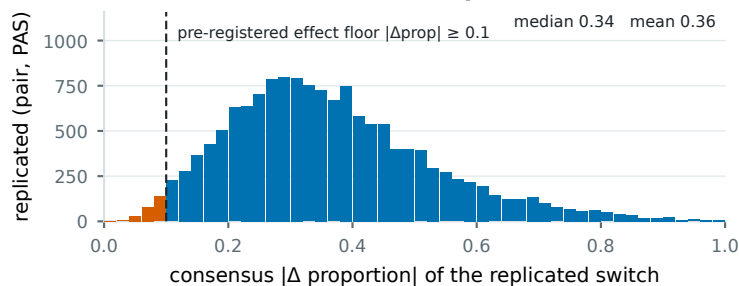
a Replication funnel, real vs patient-wise label-shuffle null (one axis)



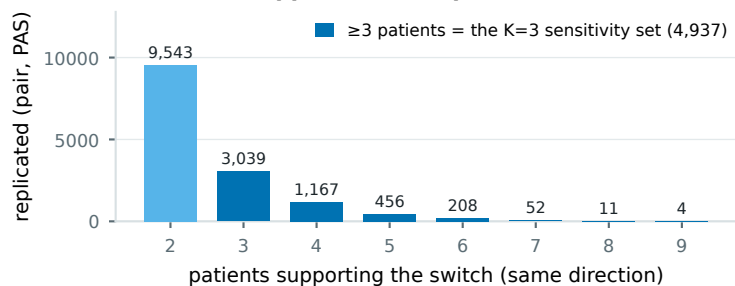
b Top 16 of the 47 cell-type pairs tested in ≥ 2 patients ('n pt' = patients tested)



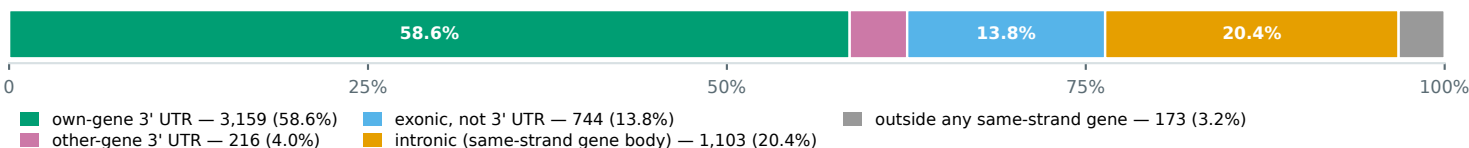
c Effect size of the 14,480 replicated switches



d Patient support of the replicated set



e Genomic context of the 5,395 replicated PAS (recomputed here from the Ensembl GRCh38.99 GTF)



Exclusive, strand-matched partition of the distinct replicated PAS. Roll-ups: 96.8% of the replicated PAS sit inside a same-strand gene body, 76.3% are exonic, 62.6% are in some gene's 3' UTR, 20.4% are intronic and 3.2% fall outside any same-strand gene (of these, only 62.5% and 96.8% appear in 20 disclosure 1, both reproduced here to within 0.1 pp; the rest are recomputed for this figure). BUT only 58.6% lie in the 3' UTR of the gene the caller ASSIGNED them to: 216 of the 3,375 3'-UTR PAS (6.4%) lie in a DIFFERENT gene's 3' UTR, and 497 (9.2%) lie inside ≥ 2 overlapping same-strand gene models. NO ranked list of named top-switch genes is shown here: 33% of the top-30 gene-level rows in 20 name a spanning/readthrough model rather than the gene whose 3' UTR holds the PAS, and seven of them have zero assigned PAS cohort-wide (tool issue #99). Replication statistics are unaffected — they are computed on PAS identifiers, never on gene names.

Design: one cohort run so all 15 libraries share a single PAS identifier space; pre-registered precision-first universe (IP-filtered $\wedge$ tier-1 $\wedge$ cohort clip-molecule sum ≥ 2) = 74,954 PAS; 29,063 curated cells $\rightarrow$ 18,651 confirmed (64.2%) by the pre-registered label policy (20), of which 17,373 entered the tests in the 12 primary patients; 59 cell-type pairs; test = Fisher on cells with no marker pre-selection (14); unit of replication = patient, so the 3 patients with 2 libraries each count once. RECORD: PeakATail code 4efeb125 (pre-Stage-1d) $\rightarrow$ v1-code record; the Stage-3 v2 chain on code 9dfdefb (stage3_laughney_v3) is running and will supersede these numbers (re-run with LAUGHNEY_SWITCHES_VERSION=v2_code); source of truth manuscript/20_stage3_replication.md (verifier verdict FIXED). NULL: not one feature replicates in ANY of the 10 patient-wise label-shuffle combinations run through the identical pipeline: 0 replicated in every one of the 10, over a comparable 1,888,790 (pair, PAS) hypotheses tested per combination (54.4M null tests produced 10 nominal $q < 0.05$ calls, none co-occurring in two patients). Ten permutations resolve only to an empirical $p \leq 0.091$; report this as 'none in 10 nulls', NEVER as an FDR estimate. EXCLUSION: GSM3516664-MetBone is excluded by pre-registration (13 addendum item 8), but its stated premise was WRONG — the 0.015% clip rate came from the caller sampling only the first 200,000 CB reads of a coordinate-sorted BAM (head of chr1); MetBone in fact has 306k clip molecules (tool issue #99). Never repeat the 'no clip evidence' justification. The 13-patient sensitivity run that includes it gives 14,541 / 13,888, a 0.4% difference confined to one pair. GENE NAMES: no named top-gene list may be published until PAS $\rightarrow$ gene re-assignment (panel e; 20 disclosure 1, tool issue #99) — the replication statistics themselves are computed on PAS ids. CAVEATS: one cohort, one chemistry, one caller — no orthogonal 3'-end assay confirms these sites; pairs tested in more patients replicate more, so panel b tracks cohort composition as much as biology; one library (GSM3516672) trips the per-GSM null rule only through BH discreteness (3 hits in 8.4M null tests) and is kept and disclosed (20). Panel c plots the consensus effect, but the pre-registered floor is applied PER PATIENT and the patients re-counted, so it removes 654 switches (14,480 $\rightarrow$ 13,826) — more than the 250 whose consensus alone falls below 0.1. Every plotted value: results/figures/manuscript/laughney_switches*.tsv.